

Using

$$\tilde{f}(k) = \int_{-\infty}^{\infty} f(x) e^{-ikx} dx, \quad f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \tilde{f}(k) e^{ikx} dk,$$

here are five basic Fourier-transform problems.

1. Exponential decay

Find the Fourier transform of

$$f(x) = e^{-a|x|}, \quad a > 0.$$

2. Gaussian

Find the Fourier transform of

$$f(x) = e^{-ax^2}, \quad a > 0.$$

You may use

$$\int_{-\infty}^{\infty} e^{-ax^2} dx = \sqrt{\frac{\pi}{a}}.$$

3. Rectangular pulse

Find the Fourier transform of

$$f(x) = \begin{cases} 1, & |x| \leq L, \\ 0, & |x| > L. \end{cases}$$

Also evaluate $\tilde{f}(0)$ using the limit $k \rightarrow 0$.

4. Shifted function

Suppose

$$g(x) = f(x - x_0).$$

Starting from the definition of the Fourier transform, show that

$$\tilde{g}(k) = e^{-ikx_0} \tilde{f}(k).$$

Then apply this result to the shifted rectangular pulse

$$g(x) = \begin{cases} 1, & |x - x_0| \leq L, \\ 0, & |x - x_0| > L. \end{cases}$$

5. Derivative property

Assume that $f(x) \rightarrow 0$ as $x \rightarrow \pm\infty$. Use integration by parts to show that

$$\mathcal{F}\left[\frac{df}{dx}\right](k) = ik\tilde{f}(k).$$

Then use this property to find the Fourier transform of

$$f(x) = \frac{d}{dx}e^{-ax^2}.$$